

Akhil Babu Nambur

Computer Vision | Agentic AI | AI/ML

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PROFESSIONAL SUMMARY

AI engineer with 2+ years at Accenture building computer vision and LLM systems for engineering documents. On the vision side: visual change detection for engineering drawings (OpenCV, NumPy) that cut manual inspection effort by 70%, torque-value extraction with Tesseract OCR and Hough-circle detection, and SAM + VGG16 callout matching that improved accuracy by 25%. Also built multi-agent LLM applications (LangChain, LangGraph, FastAPI) that run these tools in production. Published an SCI-indexed paper on privacy-preserving vessel detection with an extended YOLOv7, and co-inventor on a published international (PCT) patent application for an aerial intruder-detection vehicle.

PROFESSIONAL EXPERIENCE

Accenture

Full Stack LLM Development Associate

Bengaluru, India

Sep 2024 – Present

- Engineered an automated visual comparison pipeline with OpenCV and NumPy to detect added, removed, and shifted elements across engineering drawings, cutting manual engineering inspection effort by 70% through algorithmic difference mapping and structural verification workflows.
- Built a production multi-agent engineering AI chatbot using LangChain and LangGraph, implementing 5+ specialized agents with Supervisor-based routing and reducing manual tool switching by 40%.
- Automated torque-value extraction from engineering design sheets using Tesseract OCR, OpenCV Hough-circle callout detection and regex, linking each torque value to its Bill of Materials entries.
- Built offline desktop CV tools (Flask backend packaged with PyInstaller, Electron/Angular UI), locked to each user's machine with SHA-256 hostname-based licence keys.

Accenture IndustryX

Computer Vision & AI Intern

Bengaluru, India

Feb 2024 – Jun 2024

- Built a computer vision pipeline to match part callouts between old and new revisions of aerospace technical illustrations, where the same part has different callout numbers and is partly redrawn.
- Used the Segment Anything Model (SAM) with point and box prompts to segment the exact component each callout points to, replacing contour-based extraction that failed on touching parts in assembly drawings.
- Extracted ImageNet-pretrained VGG16 features (4096-d) for each segmented part and ranked old-vs-new candidate matches by cosine similarity.
- Combined extracted visual embeddings with structured Mistral prompts, driving a 25% improvement in comparison accuracy over baseline methods.

PUBLICATIONS & PATENTS

- Journal Publication:** “An expert system for privacy-preserving vessel detection leveraging optimized Extended-YOLOv7 and SHA-256,” *Journal of Network and Computer Applications* (Elsevier, SCI-indexed), Vol. 238, Art. 104139, June 2025.
- International Patent:** Co-Inventor for “Ornithopter Unmanned Remote Control Aerial Vehicle for Intruder Detection and Data Preservation,” World Intellectual Property Organization (WIPO), Publication No. WO2025032370, Feb 2025.

TECHNICAL PROJECTS

DataMind: Enterprise NLP-to-SQL AI System

Personal Project

Tech Stack: Python, Mistral 7B, ChromaDB, NetworkX, SQLAlchemy, SQLite

- Built a RAG-powered natural language query engine to generate structured SQL queries over complex relational databases.
- Applied NetworkX graph algorithms for automatic table JOIN resolution, semantic caching, and strict SELECT-only security gates.

Maestro: Local Multi-Agent AI Assistant

Personal Project

Tech Stack: Python, LangGraph, LangChain, FastAPI, Ollama (Mistral), Groq (Llama 3.1), Docker

- Built a supervisor agent in LangGraph that routes each request to one of four specialist ReAct agents (math, weather, research, writing), each using its own tools, such as a safe AST-based calculator and a live weather API..
- Streamed each routing and agent step to a web UI over NDJSON with FastAPI, and made the LLM backend switchable between local Ollama and Groq cloud through configuration.

SKILLS

- **Languages:** Python, HTML, CSS, SQL
- **Computer Vision:** OpenCV, NumPy, YOLOv7, Segment Anything Model (SAM), VGG, Image Segmentation, Visual Change Detection
- **AI & NLP:** Agentic AI, Large Language Models (Mistral), RAG Architectures, LangChain, LangGraph, spaCy, Prompt Engineering
- **Backend & Databases:** FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, SQL
- **Deployment & Tools:** Docker, Linux, Git, PyInstaller, Ollama

EDUCATION

Velagapudi Ramakrishna Siddhartha Engineering College (VRSEC)

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.91 / 10.0

Vijayawada, India

2020 – 2024